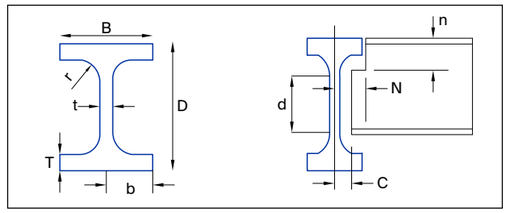
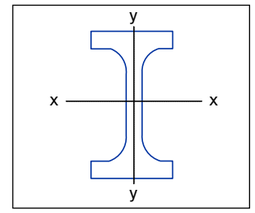
2.4
Metric Beams & Columns (MB & MC)



Dimensions	JIS G 3192
Specification	EN10025/ ASTM A36/ ASTM A572/ JIS G 3101
Size Range	MB: 100mm x 50mm to 900mm x 300mm MC: 100mm x 100mm to 400mm x 400mm



Metric Units

Designation		Depth of Section	Width of Section	Thickness		Root Radius	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing		
Size	Mass per Metre			Flange	Web				Flange	Web	End clearance	Notch	
		D	B	T	t	r	d	A	b/T	d/t	C	N	n
in (mm)	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
600x200	79.0	592	197	13	8	22	522	101	7.58	65.3	6	105	35
	92.5	596	199	15	10	13	540	118	6.63	54.0	7	105	28
	94.6	596	199	15	10	22	522	120	6.63	52.2	7	105	37
	103	600	200	17	11	13	540	132	5.88	49.1	8	105	30
	106	600	200	17	11	22	522	134	5.88	47.5	8	105	39
	118	606	201	20	12	13	540	150	5.03	45.0	8	105	33
	120	606	201	20	12	22	522	152	5.03	43.5	8	105	42
	134	612	202	23	13	22	522	171	4.39	40.2	9	105	45
600x300	133	582	300	17	12	13	522	169	8.82	43.5	8	154	30
	137	582	300	17	12	28	492	174	8.82	41.0	8	154	45
	147	588	300	20	12	13	522	187	7.50	43.5	8	154	33
	151	588	300	20	12	28	492	192	7.50	41.0	8	154	48
	170	594	302	23	14	13	522	217	6.57	37.3	9	154	36
	175	594	302	23	14	28	492	222	6.57	35.1	9	154	51
	203	602	304	27	16	28	492	259	5.63	30.8	10	154	55
	217	608	304	30	16	28	492	277	5.07	30.8	10	154	58
700x300	166	692	300	20	13	28	596	211	7.50	45.8	9	154	48
	182	700	300	24	13	18	616	232	6.25	47.4	9	154	42
	185	700	300	24	13	28	596	235	6.25	45.8	9	154	52
	215	708	302	28	15	28	596	274	5.39	39.7	10	154	56
800x300	191	792	300	22	14	28	692	243	6.82	49.4	9	153	50
	207	800	300	26	14	18	712	264	5.77	50.9	9	153	44
	210	800	300	26	14	28	692	267	5.77	49.4	9	153	54
	241	808	302	30	16	28	692	308	5.03	43.3	10	153	58
	267	816	303	34	17	28	692	340	4.46	40.7	11	153	62
900x300	210	890	299	23	15	18	808	267	6.50	53.9	10	152	41
	213	890	299	23	15	28	788	271	6.50	52.5	10	152	51
	240	900	300	28	16	18	808	306	5.36	50.5	10	152	46
	243	900	300	28	16	28	788	310	5.36	49.3	10	152	56
	283	912	302	34	18	18	808	360	4.44	44.9	11	152	52
	286	912	302	34	18	28	788	364	4.44	43.8	11	152	62
	304	918	303	37	19	18	808	387	4.09	42.5	12	152	55
	307	918	303	37	19	28	788	391	4.09	41.5	12	152	65

Surface Area per Metre	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y				
									u	x	H	J
m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³			dm⁶	cm⁴
1.92	58240	1665	24.1	4.07	1968	169	2239	265	0.870	45.2	1.40	53.0
1.95	66640	1976	23.8	4.10	2236	199	2576	312	0.861	42.9	1.67	69.2
1.93	68720	1982	23.9	4.06	2306	199	2651	315	0.864	39.7	1.67	82.4
1.96	75560	2274	24.0	4.16	2519	227	2904	358	0.863	38.4	1.93	97.3
1.94	77630	2280	24.0	4.12	2588	228	2979	361	0.865	36.0	1.94	113
1.97	88320	2716	24.3	4.26	2915	270	3357	426	0.868	33.4	2.33	147
1.95	90400	2723	24.3	4.23	2983	271	3432	429	0.870	31.7	2.34	167
1.97	103500	3178	24.6	4.32	3382	315	3893	498	0.873	28.3	2.76	237
2.32	98950	7659	24.2	6.73	3400	511	3822	786	0.876	35.5	6.11	137
2.29	102700	7675	24.3	6.63	3530	512	3963	793	0.879	32.1	6.13	173
2.33	114400	9009	24.7	6.94	3889	601	4348	921	0.886	31.2	7.27	199
2.30	118100	9025	24.8	6.85	4018	602	4489	928	0.888	28.8	7.28	241
2.35	133600	10570	24.8	6.98	4497	700	5057	1077	0.885	27.3	8.62	304
2.32	137300	10590	24.9	6.90	4624	701	5197	1085	0.887	25.5	8.63	356
2.34	162600	12680	25.1	7.00	5401	834	6101	1292	0.888	22.3	10.5	551
2.35	179300	14090	25.4	7.13	5896	927	6653	1431	0.893	20.5	11.8	702
2.51	172400	9030	28.6	6.53	4984	602	5629	936	0.878	34.3	10.2	260
2.54	197500	10820	29.2	6.83	5643	721	6338	1110	0.887	31.4	12.4	343
2.53	201500	10830	29.3	6.78	5757	722	6464	1116	0.889	30.0	12.4	383
2.55	237100	12890	29.4	6.86	6699	854	7559	1323	0.889	26.3	14.9	588
2.71	253600	9936	32.3	6.39	6405	662	7288	1036	0.873	36.8	14.7	341
2.74	286400	11720	33.0	6.67	7159	781	8098	1210	0.882	33.9	17.6	440
2.72	291700	11740	33.0	6.62	7292	782	8243	1216	0.883	32.5	17.6	486
2.74	339200	13820	33.2	6.70	8397	915	9534	1426	0.883	28.7	20.9	726
2.76	383600	15820	33.6	6.82	9402	1044	10680	1625	0.887	25.8	24.2	1003
2.92	338500	10280	35.6	6.20	7608	687	8750	1079	0.863	42.2	19.3	361
2.90	345300	10290	35.7	6.16	7760	688	8913	1085	0.865	40.2	19.3	403
2.94	404500	12630	36.4	6.43	8989	842	10290	1317	0.872	35.9	24.0	580
2.92	411300	12650	36.4	6.39	9139	843	10450	1324	0.873	34.5	24.0	633
2.97	491000	15660	36.9	6.59	10770	1037	12340	1622	0.876	30.1	30.2	979
2.95	497800	15670	37.0	6.56	10920	1038	12500	1629	0.877	29.3	30.2	1050
2.98	535400	17210	37.2	6.67	11660	1136	13380	1778	0.878	27.9	33.4	1234
2.96	542200	17230	37.2	6.63	11810	1137	13540	1785	0.879	27.2	33.4	1316